INCH-POUND

MIL-F-43846C
22 September 1993
SUPERSEDING
MIL-F-43846B
4 December 1975

MILITARY SPECIFICATION

FOOD PACKET, LONG RANGE PATROL, ASSEMBLY OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the assembly requirements for the Food Packet, Long Range Patrol.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

A-A-20041 - Salt, Table, Iodized
A-A-20135 - Sugar, Refined and Sugar, Brown
A-A-20175 - Chewing Gum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8970

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A-A-20177 - Candy and Chocolate Confections
A-A-20183 - Tea, Instant
L-F-560 - Fork, Knife, and Spoon, Picnic, Plastic
L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin
EE-M-101 - Matches, Safety: and Matches, Nonsafety (Strike-Anywhere)
QQ-A-1876 - Aluminum Foil
HHH-C-575 - Coffee, Instant
MMM-A-250 - Adhesive, Water-Resistant (for Closure of Fiberboard Boxes)
PPP-B-636 - Boxes, Shipping, Fiberboard

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MIL-C-3031 - Cocoa Beverage Powder
MIL-B-35023 - Beverage Base, Powdered - (Operational Ration Component)
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43135 - Chicken and Chicken Products, Cooked, Dehydrated
MIL-S-43275 - Spaghetti with Meat Sauce, Cooked, Dehydrated
MIL-C-43287 - Chili Con Carne With Beans, Cooked, Dehydrated
MIL-C-43289 - Chicken with Rice, Cooked, Dehydrated
MIL-C-43338 - Cream Substitute, Dry, Non-Dairy
MIL-B-43404 - Beef Stew, Cooked, Dehydrated
MIL-E-43749 - Escalloped Potatoes and Pork, Cooked, Dehydrated
MIL-B-43750 - Beef with Rice, Cooked, Dehydrated
MIL-P-43988 - Paper, Toilet, Tissue (Packet in Sheet Form)
MIL-C-44072 - Cookies, Oatmeal; and Brownies; Chocolate Covered
MIL-B-44135 - Beverage Base, Powder, Orange, Fortified (Operational Ration Component)
MIL-O-44136 - Oatmeal Cookie and Granola Bars (Operational Ration Component)
MIL-C-44137 - Chicken a la King, Cooked, Dehydrated
MIL-F-44279 - Fruit Bars (Operational Ration Component)
MIL-B-44281 - Beverage Base, Powder, Apple Cider (Operational Ration Component)
MIL-C-44389 - Cereal Bars (Operational Ration Component)

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

13-1-0186 - Food Packet, Long Range Patrol, Accepted Meal Bag Packing Configuration, Figure 1
13-1-0187 - Food Packet, Long Range Patrol, Accepted Pallet Pattern, Figure 2

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-UXD, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505 - Density of Plastics by Density Gradient Technique
F 88 - Seal Strength of Flexible Barrier Materials

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Components.

3.1.1 Meal bag. The meal bag shall be made from polyethylene tubing that is approved for food use and is average gauge 0.010 to 0.012-inch thick, with a density of 0.910 to 0.929 g/cm^3. No area of the bag, including gel-induced depressions, shall be less than 0.010-inch thick. The color of the polyethylene shall be in the range of numbers 34079 through 34102, or 24052 through 24087, or 30045 through 30118 (excluding 30109) or shall conform to number 10045, 20040, 20045, 20062, or 34052 of FED-STD-595. The inside width of the bag shall be not less than 7 inches and not more than 7-5/8 inches and the inside length shall be not less than 11-7/8 inches (± 1/8 inch). The bag opening shall be across the width. The bag manufacturer's seal shall be a minimum 1/16 inch wide and located with the outside edge of the seal 1/2 inch (± 1/16 inch) from one end of the bag. The seal shall form a hermetic closure. The average seal strength of each seal shall be not less than 9 pounds per inch of width and no individual specimen shall be less than 7 pounds (see 4.2.1.2). A 1/4 to 3/8 inch deep slit or V-shaped tear notch shall be located at one end of the bag within 1/2 inch of the longitudinal center line of the bag.

3.1.1.1 Meal bag marking. Each meal bag shall be printed on at least one face in permanent ink in clear black letters not less than 3/16 inch high, as follows:

FOOD PACKET, LONG RANGE PATROL

MENU NO. (1 through 8, as applicable)

(named of applicable meat component
as listed in table I component column)
(name and address of assembler)

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See menu component lists in table III. Printing of component sizes is not required and commercial names for candies are acceptable. Continuous printing is acceptable provided that the complete markings appear uninterrupted on at least one face of the bag.

3.1.2 Accessory packet. The accessory packet shall be fabricated in accordance with 3.1.2.1 or 3.1.2.2.

3.1.2.1 Preformed packets. The preformed accessory packet shall be fabricated from material made by hot-extruding or laminating 0.0015-inch thick (± 10 percent) polyethylene on 0.00035-inch thick (± 10 percent) aluminum foil and laminating the opposite side of the foil to 0.0005-inch thick (± 20 percent) polyester. The color of the exterior (polyester) side of the laminate shall conform to number 30045, 30099, 30108, 30111, 30117, 30118, 34052, 24064, 34079, 34086, or 34087 of FED-STD-595. Any difference in gloss caused by the polyester exterior layer will be considered acceptable. The inside width of the packet shall be not less than 3-5/8 inches and not more than 4 inches. The inside length of the packet shall be not less than 5-5/8 inches and not more than 6 inches. The dimensions selected from the allowable ranges shall be sufficient to contain all the components to effect compatibility with the equipment used for packet fabricating, filling, and sealing equipment. Each packet shall be heat sealed on three edges with 3/8 inch wide (± 1/8 inch) seals. The open end of the pouch shall be across the width. A 1/8 inch wide (± 1/16 inch) lip may be incorporated at the open end of the packet to facilitate opening and filling of the packet. A tear nick or notch shall be located on the outer edge of one or both side seals not more than 2 inches from the open end of the packet. Each tear nick or notch shall be a minimum 1/16 inch deep, but shall not extend into the side seal to a depth that reduces the seal width to less than 1/8 inch. The closure seal shall not be less than 1/8 inch wide located above the nick or notch. The average seal strength of the packet seals shall be not less than 3.5 pounds per inch of width and no individual specimen shall have a seal strength of less than 3.0 pounds per inch of width (see 4.2.1.2).

3.1.2.2 Form-fill-seal packet. The form-fill-seal accessory packet shall consist of a tray-shaped body with a tray-shaped or flat sheet heat sealable cover. The tray-shaped body and tray-shaped cover (when used) shall be formed by drawing flexible laminated material into an appropriately shaped cavity. The tray-shaped or flat sheet cover shall be applied by a continuous hermetic heat seal along the entire perimeter of the tray-shaped body. The tray-shaped body and tray-shaped cover (when used) of each packet shall be fabricated from 0.0015-inch thick linear low density polyethylene bonded to 0.0005-inch thick aluminum foil with 7 pound per ream low density polyethylene extrusion resin and 0.00075-inch thick oriented polypropylene bonded to the opposite side of the aluminum foil with 10 pound per ream low density polyethylene extrusion resin. The flat sheet cover (when used) of each packet shall be fabricated from 0.0015-inch thick linear low density

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polyethylene bonded to 0.00035-inch thick aluminum foil with 7 pound per ream low density polyethylene extrusion resin and 0.0005-inch thick polyester bonded to the opposite side of the aluminum foil with 10 pound per ream low density polyethylene extrusion resin. The linear low density polyethylene shall be the copolymer of ethylene and octene-1, having melt index range of 0.8 to 1.2 g/10 minutes in accordance with ASTM D 1238 and a density range of 0.918 to 0.922 g/cc in accordance with ASTM D 1505. The color of the exterior side of the laminates, (polypropylene side of the tray-shaped body and tray-shaped cover and the polyester side of the flat sheet cover) shall be as specified in 3.1.2.1. The outside dimensions of each packet shall be 5-1/2 inches wide by 6 inches long (± 1/16 inch). The closure seal at the perimeter of the packet body and cover shall be a minimum 1/4 inch wide. A tear nick or notch shall be located on one or both outside edges of the packet length or width to effect easy opening of the packet in the machine direction of the packet body and cover laminates. The tear nick or notch shall be located 3/4 inch (± 1/16 inch) from the outside edge of the end or side of the packet. Each tear nick or notch shall be a minimum 1/16 inch deep, but shall not extend into the seal to a depth that reduces the seal width to less than 1/8 inch. Alternatively, if the packet has serrated edges, the serrations may be used as tear notches provided that the serrations are located to effect easy opening in the machine direction of the packet laminates; the serrations are sharp (no plastic tailings exist); and the serration depth and minimum seal width at the serrations are as specified for the tear nick or notch. The average seal strength of the packet seals shall be not less than 3.5 pounds per inch of width, and no individual specimen shall have a seal strength of less than 3.0 pounds per inch of width when tested in accordance with 4.2.2.2. Alternatively, the form-fill-seal accessory packet may be made on equipment that does not change the shape of the bag laminate material from the flat sheet form. The body shall not be formed into a tray, it shall be in the form of a flat sheet like the cover. When this equipment is employed, the packet shall be made in accordance with the requirements of 3.1.2.1, except that all four sides of the packet bag shall be sealed in the sequence required by the equipment used. Projection of the packet bag material beyond the outside edges of the seals is permissible. Alternatively, the form-fill-seal accessory packet may be constructed as a three seamed pillow pouch, utilizing a back seam which joins the opposite edges of the packaging materials into a tube. The back seam shall be a fin seal which is formed by heat sealing the two opposite inside edges to each other and folding the fin seal flat, in the plane of the material. The tube may be formed around the contents being packed or it may be formed empty and filled in a subsequent step. The material specified in 3.1.2.1 shall be used. Tear nicks, notches, or serrations shall be provided on one or both end seals.

3.1.2.3 Accessory packet marking. Each packet shall be printed on one or both sides in permanent, black ink, in clear black letters as follows:

ACCESSORY PACKET
(letters not less than 1/4 inch high)
(Name and address of assembler)

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3.1.3 Candy packets. Packets for overwrapping of candy components shall be fabricated in accordance with 3.1.3.1 or 3.1.3.2.

3.1.3.1 Preformed packet. The preformed candy packet shall be fabricated in accordance with 3.1.2.1 except the inside width of the packet shall be not less than 1-1/2 inches or greater than 3-1/4 inches. The inside length of the packet shall be not less than 4-3/4 inches or greater than 7 inches.

3.1.3.2 Form-fill-seal packet. The form-fill-seal candy packet shall be fabricated in accordance with 3.1.2.2 except the outside dimensions of each packet shall be 5-1/4 inches wide by 7-1/4 inches long (± 1/16 inch).

3.1.3.3 Candy packet marking. Commercial names may be used as appropriate. Each packet shall be printed on one or both sides in permanent, black ink, in clear black letters as follows:

CANDY

NAME OF PRODUCT
(letters not less than 1/4 inch high)
(Name and address of assembler)

3.1.4 Menu components. The components required for menu assembly shall be as listed in table I.

TABLE I. Menu components

Component	Document reference
Beef stew	MIL-B-43404
Chicken a la king	MIL-C-44137
Chicken stew	MIL-C-43135
Chicken with rice	MIL-C-43289
Chili con carne	MIL-C-44328
Escalloped potatoes and pork	MIL-E-43749
Beef with rice	MIL-B-43750
Spaghetti with meat sauce	MIL-S-43275
Oatmeal cookie bar	MIL-O-44136
Granola bar	MIL-0-44136
Bran flake cereal bar	MIL-C-44389
Corn flake cereal bar	MIL-C-44389
Shredded wheat cereal bar	MIL-C-44389
Chocolate covered brownie 1/	MIL-C-44072
Chocolate covered oatmeal cookie 1/	MIL-C-44072
Beverage base, apple cider	MIL-B-44281
Beverage base, orange	MIL-B-44135
Cocoa beverage powder	MIL-C-3031
Fig bar	MIL-F-44279
Tea, instant	A-A-20183, type III, class B

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TABLE I. Menu components (cont'd)

Component	Document reference
Beverage base, powdered 2/	MIL-B-35023, flavors 2 through 6
Candy 3/	
Pan-coated, chocolate disks	A-A-20177, type VI
Hard candy fruit tablets	A-A-20177, type IV
Milk chocolate bar, heat resistant	A-A-20177, type VIII,
Caramels; vanilla flavor	A-A-20177, type I
Toffee rolls; chocolate flavor	A-A-20177, type II
Starch jelly candy	A-A-20177, type IX
Accessory packet	3.1.5
Spoon	L-F-560, type 4, item 13

1/ When cookies or brownies are supplied to the ration assembler in bulk packs (see 6.1), one brownie or one cookie shall be unit packed in accordance with the applicable requirements of MIL-C-44072. Bulk packed brownies and cookies shall not be stored above 80°F and if storage time exceeds 30 days, they shall be stored at 0°F or below. Frozen cookies or brownies shall be tempered for approximately 24 hours at 65° ± 5°F and not more than 55 percent relative humidity prior to insertion into cookie/brownie bags.

2/ To be procured in equal quantities and assembled into appropriate menus in as uniform a distribution as possible in each shipping container.

3/ Candy shall be overwrapped in accordance with 3.3.

3.1.5 Accessory packet components. The components required for accessory packet assembly shall be as listed in table II.

TABLE II. Accessory packet components

Component	Document reference	Quantity
Coffee, instant	HHH-C-575, type III, style A, blend (1), composition 1	1 envelope
Cream substitute	MIL-C-43338	1 envelope
Sugar	A-A-20135, type I, class 1, 6-gram envelope	1 envelope
Salt	A-A-20041, iodized, 4-gram envelope	1 envelope

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TABLE II. Accessory packet components (cont'd)

Component	Document reference	Quantity
Chewing gum	A-A-20175, type I, flavors 1 and 2	1 packet (2 tablets)
Matches	EE-M-101, type II, class 2	1 book
Toilet tissue	MIL-P-43988	2 packets

3.2 Accessory packet assembly. The required components indicated in table II shall be inserted in an accessory packet as specified in 3.1.2. The maximum thickness of the filled and sealed accessory packet shall be 7/8 inch. When a preformed packet is used (see 3.1.2.1), a minimum of 1/8 inch wide heat seal shall be applied to effect closure of the packet. When a form-fill-seal packet is used (see 3.1.2.2), components shall be placed in the tray-shaped or flat sheet body and the cover applied by heat sealing. The sealed accessory packets shall not show any evidence of material degradation or delamination from packet fabrication, forming, or heat sealing. The average seal strength of the packet seals shall be not less than 3.5 pounds per inch of width and no individual specimen shall have a seal strength of less than 3.0 pounds per inch width when tested in accordance with 4.2.2.2.

3.3 Candy packet filling and sealing. The required candy component shall be inserted in a candy packet as specified in 3.1.3. Excess air entrapment shall be avoided during closure sealing. When a preformed packet is used (see 3.1.3.1), the closure seal shall be located above the tear nicks or notches. When a form-fill-seal packet is used, components shall be placed in the tray-shaped body and the cover applied by heat sealing (see 3.1.3.2). The sealed candy packet shall not show any evidence of material degradation or delamination from packet fabrication, forming, or heat sealing. The average seal strength of the packet seals shall be not less than 3.5 pounds per inch of width and no individual specimen shall have a seal strength of less than 3.0 pounds per inch width when tested in accordance with 4.2.2.2.

3.4 Menu assembly. One each of the applicable components indicated in table III shall be inserted in the meal bag specified in 3.1.1. The components shall be arranged in such a manner as to occupy the minimum amount of space while maintaining a flat configuration. Excess air shall be expelled prior to closure sealing to allow packing as specified in 5.1 without damaging any of the meal bags or components therein. The open end of the bag shall be hermetically sealed with a heat seal of not less than 1/16 inch wide. The closure seal shall have an average seal strength of not less than 9 pounds per linear inch and no individual specimen shall have a seal strength of less than 7 pounds (see 4.2.4.2).

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TABLE III. Menu contents

Menu No., Components therein	Menu No., Components therein
Menu #1	Menu #2
Chicken stew Bran flake bar Oatmeal cookie bar Toffee rolls; chocolate flavor (four 1/2 ounce individually wrapped rolls per packet) Beverage base, apple cider Accessory packet Spoon	Beef stew Granola bar Cookie, chocolate covered Caramels; vanilla flavor (one 1-1/4 ounce package per packet) Cocoa beverage powder Accessory packet Spoon
Menu #3	Menu #4
Escalloped potatoes and pork Shredded wheat bar Fig bar Milk chocolate bar, heat resistant (two 1 ounce individually wrapped bars per packet) Beverage base, apple cider Accessory packet Spoon	Chicken ala king Bran flake bar Cookie, chocolate covered Starch jelly candy; assorted flavors (one 2 ounce package or four 16 gram individually wrapped packages per packet) Beverage base, orange Accessory packet Spoon
Menu #5	Menu #6
Chicken and rice Granola bar Brownie, chocolate covered Starch jelly candy; assorted flavors (one 2 ounce package or four 16 gram individually wrapped packages per packet) Tea, instant (2 packets) Accessory packet Spoon	Spaghetti with meat sauce Cornflake bar Oatmeal cookie bar Toffee rolls; chocolate flavored (four 1/2 ounce individually wrapped rolls per packet) Beverage base, powdered Accessory packet Spoon

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TABLE III. Menu contents (cont'd)

Menu No., Components therein	Menu No., Components therein
Menu #7	Menu #8
Chili con carne	Beef with rice
Shredded wheat bar	Cornflake bar
Brownie, chocolate covered	Fig bar
Hard candy fruit tablets (one 1 ounce package per packet)	Pan-coated, chocolate disks (one 1-1/2 ounce package per packet)
Beverage base, orange	Tea, instant (2 packets)
Accessory packet	Accessory packet
Spoon	Spoon

3.5 Plant qualification. The product shall be packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding Human Food", and the plant sanitation requirements of the appropriate Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.1.1 Meal bag and accesory packet material certification. Material listed below shall be accepted on the basis of a contractor's certification of conformance to the indicated requirements.

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Requirement	Requirement paragraph	Test procedure
Meal bag material:		
Identification	3.1.1	Laboratory evaluation
Thickness	3.1.1	Laboratory evaluation
Color	3.1.1	Visual evaluation
Accessory and candy packet material:		
Identification	3.1.2.1 and 3.1.2.2	Laboratory evaluation
Polyethylene, linear low density polyethylene, polypropylene and polyester film thickness	3.1.2.1 and 3.1.2.2	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Aluminum foil thickness	3.1.2.1 and 3.1.2.2	As specified in QQ-A-1876
Color	3.1.2.1 and 3.1.2.2	Visual evaluation

4.2.1.2 Unfilled meal bag, preformed accessory and candy packet certification. A certificate of compliance may be accepted as evidence that unfilled bags and packets conform to the requirements specified in 3.1.1, 3.1.2.1, and 3.1.3.1, respectively. When deemed necessary by the inspection agency, testing of the unfilled bags and packets for seal strength of manufacturer's seals shall be as specified in 4.2.1.2.1 and 4.2.1.2.2.

4.2.1.2.1 Unfilled preformed accessory and candy packet seal strength testing. The unfilled preformed packets shall be tested for seal strength in accordance with ASTM F 88 except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. One-half or 1 inch wide test specimens shall be cut to a length suitable for proper mounting. The lot size shall be expressed in packets. The sample unit shall be one unfilled, preformed accessory packet. The inspection level shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 10.0. Three adjacent specimens 1/2 or 1 inch wide, shall be cut from each of the three sealed sides of each packet in the

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sample. The average seal strength of each seal shall be calculated by averaging the test results of the three test specimens. The results shall be reported to the nearest 0.1 pound per inch of width. Any test specimen failing to meet the seal strength requirements specified in 3.1.2.1 and 3.1.3.1 shall be classified as a major defect. Failure to meet the average seal strength requirement shall be cause for rejection of the lot.

4.2.1.2.2 Unfilled meal bag seal strength testing. The unfilled bags shall be tested for seal strength in accordance with ASTM F 88 except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed may be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. One half or 1 inch wide test specimens shall be cut to a length suitable for proper mounting. The lot size shall be expressed in bags. The sample unit shall be one unfilled meal bag. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 1.5. Three adjacent specimens 1/2 or 1 inch wide, shall be cut from the sealed side of each bag in the sample. The results shall be reported to the nearest 0.1 pound per inch of width. The average seal strength shall be calculated by averaging the three test specimens cut from that seal. Any test specimen failing to meet the requirements of 3.1.1 shall be classified as a major defect. Failure to meet the average seal strength requirement shall be cause for rejection of the lot.

4.2.2 Accessory and candy packet inspection.

4.2.2.1 Accessory and candy packet examination. The filled and sealed preformed packets or form-fill-seal packets, shall be examined for the defects listed in table IV. The lot size shall be expressed in packets. The sample unit shall be one filled and sealed packet. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects.

TABLE IV. Filled and sealed accessory and candy packet defects

Category		Defect
Major	Minor	
101		Not clean 1/
	201	Effective heat seal closure width less than 1/16 inch at any point
	202	Thickness of packet greater than specified 2/
	203	Tear, nick, notch, or serrations missing; or do not facilitate easy opening; or assembler's seal extends into or below tear, nick, or notch

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TABLE IV. Filled and sealed accessory and candy packet defects (cont'd)

Category Major	Minor	Defect
	204	Dimensions not as specified
	205	Tear, hole, or open seal
	206	Evidence of delamination
	207	Labeling missing, incorrect, or illegible

1/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause packet damage (for example, glass, metal filings, etc.), or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential packet damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Localized dried product which affects less than 1/8 of the total surface area of one packet face, or an aggregate of scattered dried product which affects less than 1/4 of the total surface area of one packet face.

2/ Applicable to accessory packet only.

4.2.2.2 Accessory and candy packet closure seal strength testing. The filled and sealed packets shall be tested in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed may be 10 or 12 inches per minute. The test specimen length shall be suitable for proper mounting. For preformed packets, three adjacent specimens, 1/2 or 1 inch wide, shall be cut from the closure seal of each packet in the sample. For the form-fill-seal packets, three adjacent specimens, 1/2 or 1 inch wide, shall be cut from each side and each end of each packet in the sample. The average seal strength of the closure seal for the preformed packet shall be calculated by averaging the test results of the three test specimens cut from that seal. For the form-fill-seal packet, the average seal strength of each side and end seal of the packet shall be calculated by averaging the test results of the three specimens cut from each side seal or end seal. The results shall be reported to the nearest 0.1 pound per inch of width. The lot size shall be expressed in packets. The sample unit shall be one sealed packet. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 10.0. Any test specimen failing to meet the requirements of 3.2 or 3.3 shall be classified as a major defect. Failure to meet the average seal strength requirements shall be cause for rejection of the lot.

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4.2.2.3 Accessory and candy packet and contents examination. The filled and sealed packets shall be examined for the defects listed in table V (this examination may be performed on the preformed packets after filling and prior to sealing). The lot size shall be expressed in packets. The sample unit shall be one packet open or sealed. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for minor defects.

TABLE V. Accessory and candy packet and contents defects

Category		Defect
Major	Minor	
101		Missing or crushed candy
102		Foreign odor
103		Component not clean 1/
	201	Missing (except candy) or unserviceable component (for example, tear, hole, or open seam in coffee, cream substitute, sugar, or salt package; or crushed gum)

1/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause package damage (for example, glass, metal filings, etc.), or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential package damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Localized dried product which affects less than 1/8 of the total surface area of one package face, or an aggregate of scattered dried product which affects less than 1/4 of the total surface area of one package face.

4.2.3 Shipping container inspection.

4.2.3.1 Shipping container examination. Examination of the filled and closed shipping containers shall be in accordance with the examination criteria of the appendix to PPP-B-636 except that the closure examination shall be in accordance with 4.2.3.2. In addition, the following defects shall be included in the table of examination:

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Major - National stock number, item description, contract number or date of pack markings missing, incorrect, or illegible.
- Metal fasteners used in closing the box.
- Outer flaps do not completely meet, leaving an opening greater than 3/4 inch between flap ends.

Minor - Other required markings missing, incorrect, or illegible.

4.2.3.2 Flap closure testing. The closure of the four outer flaps of the shipping container shall be tested separately. A 90 degree angular bar with each leg approximately 5 inches long by 3 inches wide by 1/8 inch thick shall be used to test the flap closures. Insert one leg of the angular bar full length under the center of one outer flap. Insertion shall be made through the open slot between the outer flaps. Lift the container vertically by the other leg of the bar until the container is suspended. The complete upper surface of the inserted leg shall be in contact with the inner surface of the flap during the lifting and suspension of the container. Complete separation of the adhesive bond of one or more of the outer flaps, showing no evidence of fiber tear, shall be classified as a major defect. The lot size shall be expressed in shipping containers. The sample unit shall be one shipping container. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.3.3 Shipping container interior examination. The filled and sealed shipping containers shall be examined for the defects listed in table VI. The lot size shall be expressed in shipping containers. The sample unit shall be one filled and sealed shipping container. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for major defects and 10.0 for minor defects.

TABLE VI. Shipping container interior defects

Category		Defect
Major	Minor	
101		Less than or more than 16 menus
102		Not two of each menu
103		Box liner missing
	201	Not packed or arranged as specified

4.2.3.4 Examination of adhered sleeve joint. A certificate of compliance may be accepted as evidence that the sleeve glue joint conforms to the requirements for weather-resistant boxes, glue closure examination, as specified in the appendix of PPP-B-636.

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4.2.4 Meal bag inspection.

4.2.4.1 Meal bag examination. The filled and sealed meal bags shall be examined for the defects listed in table VII. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed meal bag. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects. The sample meal bags shall be selected from shipping containers which have been filled, sealed, sleeved, and strapped. The inspection sample shall contain a proportionate amount of each of the 8 menus.

TABLE VII. Meal bag defects

Category		Defects
Major	Minor	
101		Menu component missing or incorrect assortment for menu package
102		Foreign odor
103		Not clean - the meal bag or any of the outer packaging of its contents 1/
104		Labeling missing, incorrect, or illegible
105		Tear, hole, puncture, or open seal in packaging of menu component
106		Loss of vacuum - entrees, cookie, oatmeal cookie bar, granola bar, fig bar, cereal bar or brownie bag 2/
107		Crushed or broken cookie, oatmeal cookie bar, granola bar, fig bar, or brownie bag 3/
108		Broken spoon
	201	Tear notch or slit missing or not as specified
	202	Tear, hole, open seal, or split in meal bag
	204	Tear, hole, puncture, or open seal in accessory packet bag

1/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal filings, etc.), or generally detracts from the clean appearance of the package. The following examples shall not be scored as a defects for unclean:

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work interlock and stationary mushroom bolts mounted in the blaster.

a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Localized dried product which affects less than 1/8 of the total surface area of one pouch face, or an aggregate of scattered dried product which affects less than 1/4 of the total surface area of one pouch face.

c. Water spots.

d. Very thin film of grease, oil, or product residue which is discernible to touch, but is not readily discernible by visual examination.

2/ When vacuum retention cannot be determined visually by obvious cling of the bag walls to the contents, retention shall be verified by testing as specified in the applicable component specification.

3/ Examination for crushed product shall be performed on closed component bags. A crushed component is one in which 1/8 or more of the volume of the item has been reduced to crumbs or powder.

4.2.4.2 Meal bag closure seal strength testing. The filled and sealed meal bags shall be tested for closure seal strength in accordance with ASTM F 88 except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed may be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. One half or 1 inch wide test specimens shall be cut to a length suitable for proper mounting. Three adjacent specimens, 1/2 or 1 inch wide, shall be cut from the closure seal of each bag in the sample. The results shall be reported to the nearest 0.1 pound per inch of width for each specimen and for the average of the three specimens from each closure seal. The lot size shall be expressed in bags. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 1.5. Any test specimen failing to meet the requirements of 3.4 shall be classified as a major defect and shall be cause for rejection of the lot.

4.2.5 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Packing. Sixteen meals, two of each menu of table III shall be packed in a fiberboard box with box liner and sleeve constructed and closed in accordance with style RSC-L-SL, grade V2s of PPP-B-636, except metal fasteners shall not be used in the closure of the flaps, the gap between the outer flaps shall be not more than 3/4 inch wide, and adequacy of the closure shall be determined by testing as specified in 4.2.3.2. The inside of each

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shipping container shall be fitted with a box liner conforming to grade V3c of PPP-B-636. The height of the box liner shall be 1/4 (± 1/16) inch shorter than the inside height of the shipping container. An overlapped glued joint located at the center of a side panel or in a corner of the box liner may be used in lieu of the taped joint. The amount of overlap shall not be less than 1-1/4 inches and shall extend along the full height of the box liner. Inner flaps of the box may be closed with two strips of hot melt adhesive per flap, provided the closure meets the test requirement of 4.2.3.2. As an alternative to body joint metal fasteners, adhesive conforming to MMM-A-250, applied in accordance with "adhesive joint" requirement of PPP-B-636, may be used to secure the manufacturer's joint of the V2s box. As an alternative to sleeve joint metal fasteners, adhesive conforming to MMM-A-250 may be used for securing the joint, except the overlap shall be a minimum of 3 inches wide. The adhesive joint shall be examined in accordance with 4.2.3.4. The adhesive shall cover a minimum of 90 percent of the sleeve overlap area. Meal bags, assembled as specified in 3.4, shall be packed on edge, and arranged as specified in figure 1. Boxes shall be reinforced in accordance with the appendix of PPP-B-636, except pressure-sensitive adhesive filament reinforced tape is prohibited and boxes shall be reinforced with two girthwise straps positioned to divide the box into units of equal length. The outside dimensions of the fiberboard box shall be 24-1/2 inches in length, 16-1/4 inches in width, and 6-1/4 inches in depth.

5.2 Unit loads. The Food Packet, Long Range Patrol, packed as specified in 5.1, shall be arranged in unit loads in accordance with type I, class B of MIL-L-35078, except that the pallet pattern shall be as specified in figure 2.

5.3 Labeling and marking.

5.3.1 Shipping containers. Shipping containers for complete LRPs and those for shipping components to the ration assembler shall be marked in accordance with MIL-STD-129 and as specified in the contract with the following exceptions:

a. Identification markings normally placed on an end of the shipping container will be placed on the corresponding end of the sleeve. Contract data and other required markings will be placed on the side of the shipping container to the right of the marked end of the sleeve (when sleeve is positioned on the shipping container).

b. The top of the sleeve shall be marked "FOOD PACKET, LONG RANGE PATROL" in letters not less than 1-1/4 inches high.

c. The bottom end of the sleeve as well as the bottom of the box shall convey the box maker's compliance markings as required.

5.3.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078 and as specified in the contract.

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Step 13: Design the door supports for the positive dynamic

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

6.2 Subject term (key word) listing.

Combat field feeding
Operational rations

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, QM
Navy - MC
DP - SS

Preparing activity:

Army - GL

(Project 8970-0170)

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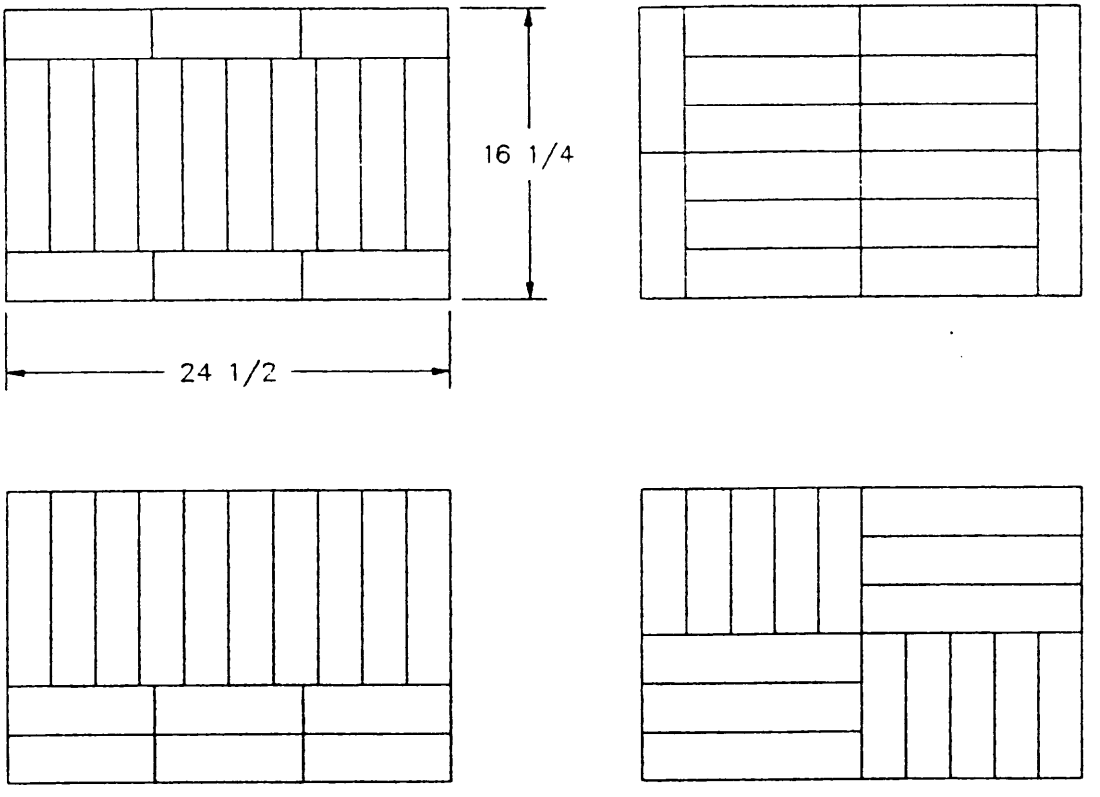


FIGURE 1. ACCEPTED MEAL BAG PACKING CONFIGURATIONS

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OUTSIDE DIMENSIONS

	LENGTH	WIDTH	DEPTH	
BOX	24.5	16.25	6.25	IN
LOAD	49.0	40.75	43.0	IN
OVERHANG	0.5	0.375	0.5	IN

CUBE = 1.44 CU FT
49.69 CU FT

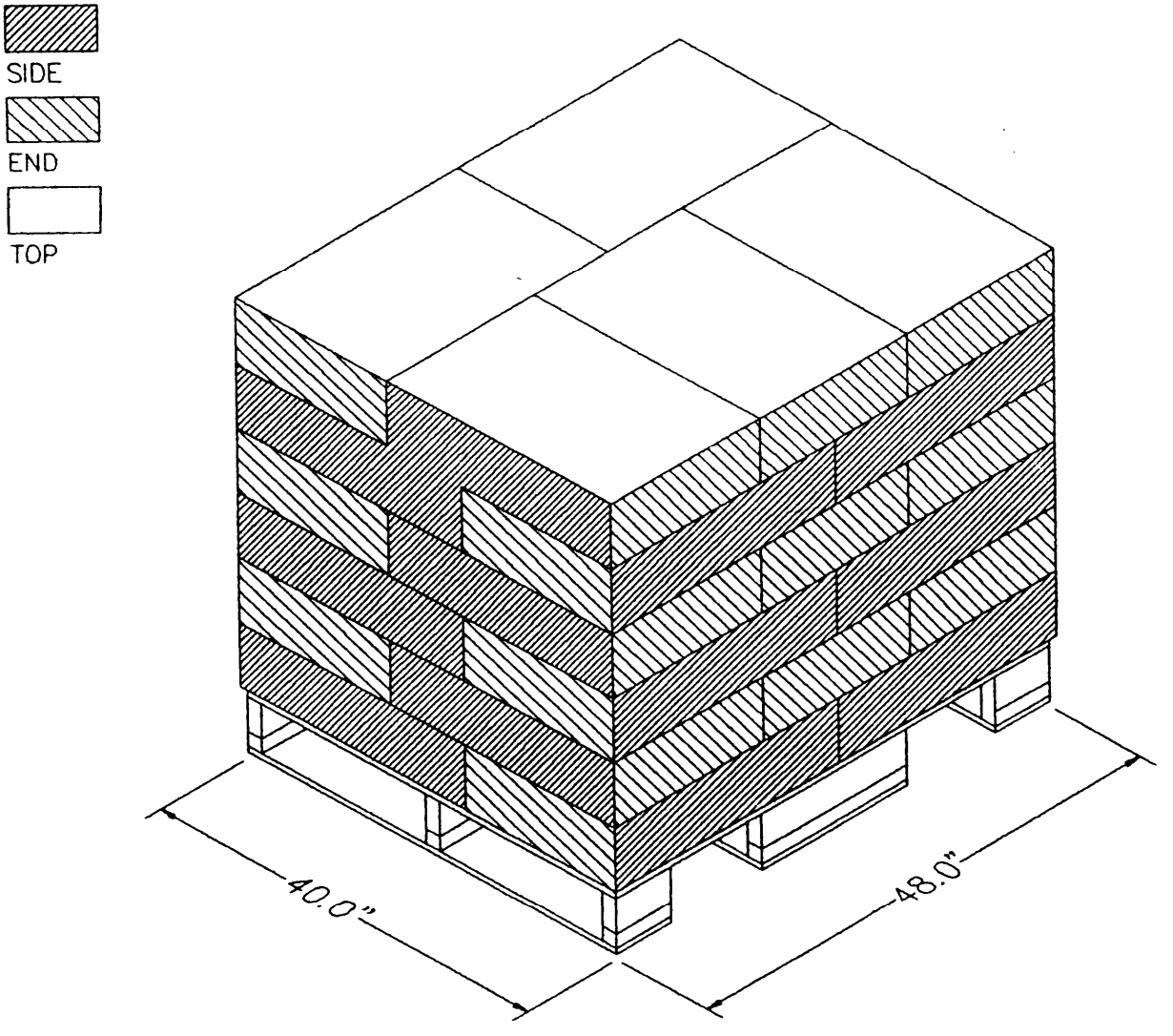


FIGURE 2. ACCEPTED PALLET PATTERN

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-43846C	1993 September 22

3. DOCUMENT TITLE

FOOD PACKET, LONG RANGE PATROL, ASSEMBLY OF

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WT Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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